

Codefest 9

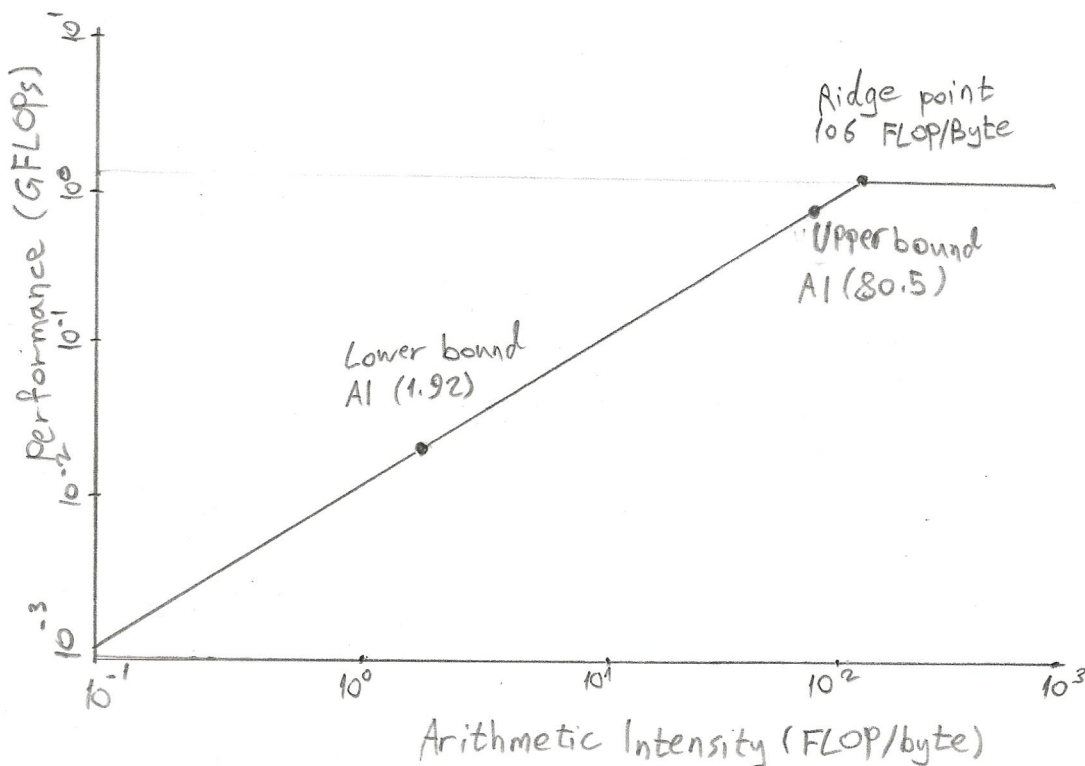
1. The dominant kernel is the dense matrix-vector-multiply (MVM) across four layers. $W[16 \times 32]$, $W[8 \times 16]$, $W[16 \times 8]$, $W[32 \times 16]$, INT8 weight and activations, and 32-bit accumulator.
2. $FLOPs = 1280 MACS * 2 = 2560 FLOPs (MVM) + 97 FLOPs (MSE) = 2657 FLOPs$ per inference
3. Lower bytes = weights + biases + input + output = $1280 + 72 + 32 + 1 = 1385$ bytes
 Lower bound AI = $FLOPs / Bytes = 2657 / 1385 = 1.92 FLOPs/Bytes$

Upper bytes = Input + outputs = $32 + 1 = 33$ bytes

Upper bound AI = $FLOPs / Bytes = 2657 / 33 = 80.5 FLOPs/byte$

The weight stationary is the reuse pattern where the weights are loaded into the chip SRAM will increase the arithmetic intensity from 1.92 to 80.5 which moves the workload from memory bound region to near the ridge point in the roofline model.

4.



5- My design is limited by interface bandwidth (AXI4-stream). The highest-leverage change I can make is widen the AXI4 interface to overlap I/O with the compute.